

Cloned Voices, Real Consequences: Evaluating Bias in Political Deepfake Detection for Electoral Integrity in Brazil

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Abstract—Recent advances in generative artificial intelligence have made it easier to fabricate statements and amplify political disinformation during elections. We introduce ParlaSpoof-BR, an audio deepfake dataset derived from recordings of the Brazilian Chamber of Deputies and expanded with synthetic utterances from diverse text-to-speech and voice conversion models. Using ParlaSpoof-BR, we benchmark state-of-the-art audio deepfake detectors, examine their ability to generalize to Brazilian Portuguese political speech, and investigate potential biases in their predictions. Our analysis reveals that current systems struggle to provide consistent decisions across the diversity represented in the dataset, with methodological factors (synthesis model choice, manipulation extent) dominating over demographic disparities. ParlaSpoof-BR provides a domain-specific benchmark for studying audio deepfake detection in a socially consequential and underrepresented setting, supporting the development of more robust detection systems for electoral integrity in Brazil.

Index Terms—audio deep fake detection, anti-spoofing, multi-media forensics

I. INTRODUCTION

Political disinformation poses a persistent threat to democratic processes by disrupting public debate, eroding trust in electoral institutions and political information, and shaping voters’ beliefs and perceptions [1]. Recent advances in Generative Artificial Intelligence (Generative AI) have intensified this challenge by lowering the cost of producing convincing false content at scale [2, 3]. Audio deepfakes are particularly concerning because they can fabricate statements in the voices of political figures, allowing false narratives to shape public opinion before journalists and fact-checkers can effectively respond [2, 3]. This risk is especially salient in Brazil, where the 2022 election was marked by coordinated disinformation campaigns [4]. The 2026 presidential election further highlighted this threat, as zero-shot voice-cloning technologies capable of generating highly realistic speech from only a few seconds of reference audio had become widely accessible.

These developments underscore the critical importance of reliable audio deepfake detection for electoral integrity.

Yet existing audio deepfake detectors provide limited assurance in realistic settings. Although they achieve near-perfect EER on ASVspoof [5], a widely used benchmark for speech anti-spoofing, their performance degrades by an order of magnitude on real-world audio. In these conditions, detectors often rely on spurious cues, such as silence statistics, bitrate signatures, and spoken content, rather than artifacts intrinsic to speech synthesis [6, 7]. Detection errors also vary across gender, age, and accent [8], while representative Portuguese-language resources remain scarce. The only dedicated corpus, BRSpeech-DF [9], consists of studio-clean audiobook recordings and does not include voice conversion, partial manipulation, or regional stratification.

In this paper, we introduce **ParlaSpoof-BR**¹, an audio deepfake benchmark based on Brazilian parliamentary speech. The dataset is publicly available for research purposes². It contains 2,000 utterances from 40 speakers balanced across gender and region, with attacks covering text-to-speech (TTS), voice conversion (VC), and semantically targeted partial manipulation. Our contributions are threefold: (1) the first Portuguese benchmark for political speech deepfakes; (2) a partial-manipulation protocol that is harder to detect than full synthesis; and (3) a systematic bias analysis showing that methodological factors have a greater impact than demographic differences.

II. RELATED WORK

Deepfakes and Elections. Synthetic media pose a growing threat to electoral integrity worldwide. Pawelec [2] analyzes how deepfakes can undermine democratic discourse by fabricating statements attributed to candidates. Twomey *et al.* [3]

¹<https://ermisai.github.io/parlaspoof-br-demo>

²<https://huggingface.co/datasets/freds0/ParlaSpoof-BR>

show that synthetic media can erode epistemic trust, meaning confidence in distinguishing reliable information from falsehoods, because their effects may occur before fact-checking can correct the record. In Brazil, the 2022 election was marked by coordinated disinformation campaigns on WhatsApp [4]. More recently, manipulated and AI-generated political audio has been documented during the 2024 elections in Brazil [10]. These studies motivate domain-specific benchmarks targeting political speech rather than generic audio.

Cross-Domain Generalization and Robustness. Audio deepfake detectors often degrade substantially outside their training domain. Müller *et al.* [11] show that systems achieving strong results on ASVspoof perform considerably worse on real-world recordings, while Speech DF Arena [12] confirms this limitation across multiple datasets and detectors. Robustness studies further show that compression, additive noise, and reverberation can suppress synthesis artifacts and increase detection errors [13, 14]. These findings motivate our evaluation of detectors on spontaneous Brazilian political speech under realistic acoustic perturbations.

Bias in Deepfake Detection. Detectors exhibit systematic biases that aggregate metrics conceal. FairSSD [8] audits six detectors over 0.9 million signals and finds most biased with respect to gender, age, and accent, with systematically higher false positive rates for male speakers. Fursule *et al.* [15] demonstrate statistically significant gender disparities that are obscured by aggregate metrics. Their follow-up study [16] shows that group-specific thresholds can reduce false-positive-rate disparities by 54–75% without reducing overall detection accuracy. Cross-lingual transfer is a persistent weakness: Liu *et al.* [17] measure language mismatch across twelve languages, and Moreno *et al.* [18] show that even holding TTS architecture fixed, detection rates vary systematically by language. To the best of our knowledge, no previous audio deepfake detection study has examined within-language regional variation as a fairness dimension; ParlaSpoof-BR addresses this gap by considering Brazil’s five geographic regions explicitly.

Portuguese Deepfake Resources. Portuguese-language resources for audio deepfake detection remain limited. BRSpeech-DF [9], introduced as the first publicly available dataset in this setting, provides over 458,000 utterances covering Brazilian and European Portuguese. Its data are derived from enhanced LibriVox audiobook recordings from 62 speakers and synthetic speech generated by five zero-shot TTS systems. Evaluations on BRSpeech-DF revealed substantial cross-lingual generalization challenges, with English-trained detectors obtaining EERs between 30.67% and 63.03%. ParlaSpoof-BR complements this large-scale resource by targeting spontaneous Brazilian political speech and expanding the attack coverage to TTS, voice conversion, and partial manipulation, with explicit regional and gender balance.

Partial Manipulation. Fully synthesizing an utterance is unnecessary when replacing a few words suffices to invert meaning. Liu *et al.* [19] show detectors concentrate on transition regions rather than synthesis artifacts, and Gajewska *et al.* [20] report 14.5% of real fraud cases involved partial

edits. Our infilling protocol extends this work by selecting targets semantically and cross-fading boundaries to suppress transition cues.

III. METHODOLOGY

A. Dataset

ParlaSpoof-BR was constructed from recordings collected between 01/01/2024 and 07/01/2026 from the official Sound Archive of the Brazilian Chamber of Deputies³. These recordings preserve the reverberation and microphone variability of parliamentary proceedings and are distributed under a Creative Commons Attribution license, allowing commercial use. Speaker identities and segment boundaries were obtained directly from the archive metadata.

The dataset comprises 40 speakers, balanced by gender (20 male and 20 female) and distributed across Brazil’s five geographic regions. Eligible speakers had at least 100 segments of five seconds or longer and 30 segments of ten seconds or longer. We selected 50 utterances per speaker, yielding 2,000 bona fide samples, and five additional segments of at least ten seconds per speaker as voice-cloning references. These references were kept separate from the bona fide evaluation samples. Each utterance includes the source waveform, an automatic transcription, and word-level timestamps obtained with WhisperX [21].

B. Audio Deepfake Generation

We employ three complementary attack paradigms: Text-to-Speech (TTS), Voice Conversion (VC), and partial manipulation through audio infilling.

Text-to-Speech. We evaluate five state-of-the-art (SOTA) zero-shot voice-cloning models capable of generating audio in Brazilian Portuguese: OmniVoice [22], XTTS-v2 [23], Chatterbox Multilingual V3 [24], VoxCPM2 [25], and Qwen3-TTS [26]. For each cross-speaker pair, speaker *A* provides the target voice and speaker *B* provides the linguistic content. Each model processes all 2,000 source utterances, yielding 10,000 TTS files.

Voice Conversion. We evaluate five zero-shot VC models: Seed-VC [27], kNN-VC [28], OpenVoice-v2 [29], X-VC [30], and EZVC [31]. For each source utterance, speaker *B*’s audio is converted to speaker *A*’s identity while preserving its linguistic content. Applying all five systems to the 2,000 source utterances yields 10,000 VC files.

Audio Infilling. We use OmniVoice [22] in masked-infilling mode to regenerate selected regions of an utterance conditioned on the surrounding audio, avoiding the need to synthesize and splice independent segments. WhisperX [21] word alignments determine the corresponding frame boundaries. We construct four conditions, yielding 8,000 partially manipulated audios. In the LLM-guided semantic condition, Claude Sonnet 4 [32] selects meaning-altering edits from five categories: ANTONYM, NUMBER, NAME, PHRASE, and NEGATION. In the three contiguous-resynthesis conditions, the original words

³<https://imagem.camara.leg.br/internet/audio/>

TABLE I
COMPOSITION OF PARLASPOOF-BR.

Label	Subset	Files
Bona fide	Original Chamber recordings	2,000
	Resemble Enhance ⁵ , Demucs [33], MetricGAN+ [34]	6,000
	MP3/OGG→WAV variants	3,200
Spoof	TTS: five systems	10,000
	Voice conversion: five systems	10,000
	OmniVoice speech infilling	8,000
	Babble noise: three SNR levels	60,000
	MP3/OGG→WAV variants	35,200
	Voice-cloning references	200
Core evaluation set		30,000
Robustness variants		104,400
Bona fide / spoof totals		11,200 / 123,200
Total benchmark files		134,400

within spans covering 25%, 50%, or 75% of the utterance are regenerated without changing the linguistic content. The former models realistic semantic manipulation, whereas the latter isolates acoustic cues introduced by partial synthesis.

C. Acoustic Robustness Conditions

Enhancement.

We process each of the 2,000 bona fide utterances with three speech enhancement systems: Resemble Enhance⁴, Demucs [33], and MetricGAN+ [34]. Producing one variant per system and 6,000 enhanced files in total.

Lossy Compression. For each codec, we sample 20% of every subset, stratified by speaker and attack method, transcode the selected files to MP3 or OGG, and decode them back to WAV. Codec-derived files retain the label and source identifier of their uncompressed counterparts. Each codec produces 19,200 variants, from which 1,600 are bona fide or bona-fide-derived files and 17,600 spoofed files, for 38,400 files in total. Voice references are excluded.

Babble Noise. Parliamentary recordings often contain overlapping speech and background conversations, producing interference similar to babble noise. To better reproduce these acoustic conditions, we construct babble signals from random speech segments extracted from other recordings using Silero VAD [35]. We add this noise to the 20,000 full-synthesis attacks, comprising 10,000 TTS and 10,000 VC files, at 20, 15, and 10 dB SNR. One variant is generated at each SNR, yielding 60,000 additional spoofed files. Babble noise is not applied to bona fide or partially manipulated speech.

D. Synthetic Speech Quality Assessment

We assess clean synthetic speech in terms of perceptual quality, intelligibility, and target-speaker similarity. UTMOS [36] estimates perceptual quality, WER and CER are computed from WhisperX [21] transcriptions, and speaker similarity is measured using the cosine similarity between ECAPA-TDNN embeddings [37]. TTS transcriptions are compared with the synthesis text, whereas VC transcriptions are compared with the corresponding source transcript. We

⁴<https://github.com/resemble-ai/resemble-enhance>

TABLE II
QUALITY, INTELLIGIBILITY, AND SPEAKER SIMILARITY FOR UNCOMPRESSED SYNTHETIC SPEECH. EACH GENERATOR CONTRIBUTES 2,000 SAMPLES. VALUES ARE MEDIANS; WER AND CER ARE PERCENTAGES.

Type	Generator	UTMOS [↑]	WER [↓]	CER [↓]	ECAPA [↑]
TTS	Chatterbox	2.983	2.6	0.7	0.802
	XTTS-v2	2.464	4.3	1.4	0.687
	OmniVoice	2.680	3.3	1.0	0.852
	Qwen3-TTS	2.980	4.0	1.3	0.843
	VoxCPM2	2.524	3.3	0.9	0.819
VC	EZ-VC	2.452	19.6	10.2	0.753
	kNN-VC	1.943	9.7	3.4	0.762
	OpenVoice-v2	2.125	7.5	2.7	0.444
	Seed-VC	1.998	7.1	2.6	0.839
	X-VC	1.714	6.2	2.4	0.791

exclude files modified by babble noise or lossy codecs to avoid conflating generation quality with acoustic degradation. Table II reports medians because the metric distributions are asymmetric. Higher UTMOS and ECAPA scores are preferred, while lower WER and CER indicate better intelligibility.

Among the TTS systems, Chatterbox achieves the highest UTMOS (2.983) and lowest error rates (2.6% WER and 0.7% CER), while OmniVoice obtains the highest speaker similarity (0.852). Qwen3-TTS also performs strongly in quality and similarity. Among the VC systems, EZ-VC achieves the highest UTMOS (2.452), Seed-VC the highest speaker similarity (0.839), and X-VC the lowest error rates (6.2% WER and 2.4% CER). Overall, TTS provides better estimated quality and intelligibility, while no system performs best across all dimensions.

E. Detection Systems

We evaluate three detectors representing different points on the capacity-generalization trade-off:

AASIST [38] is a graph-attention-based anti-spoofing system that operates on raw waveforms using a RawNet2-based [39] encoder. We use the standard pretrained configuration, trained on the ASVspoof 2019 Logical Access dataset, as an established baseline for measuring cross-domain generalization.

AASIST-L [38] is the lightweight AASIST variant, containing 85,306 parameters while retaining the same overall spectro-temporal graph-attention design. Comparing it with the higher-capacity standard AASIST configuration examines whether model capacity within this architecture is associated with improved out-of-domain performance.

DF-Arena-1B [12] is an industrial-grade transformer (1B parameters) trained on diverse multilingual deepfake corpora. It represents the best-case scenario for heterogeneous training and tests whether massive scale can overcome domain shift to Portuguese political speech.

F. Evaluation

The complete benchmark comprises 134,400 files: 11,200 bona fide or bona-fide-derived files and 123,200 spoofed files. We separate the 30,000-file core evaluation set (2,000 original bona fide files and 28,000 unperturbed TTS, VC, and infilling attacks) from the 104,400-file robustness set.

Confusion Matrices on ParlaSpoof-BR (11,248 bonafide | 114,867 spoof)

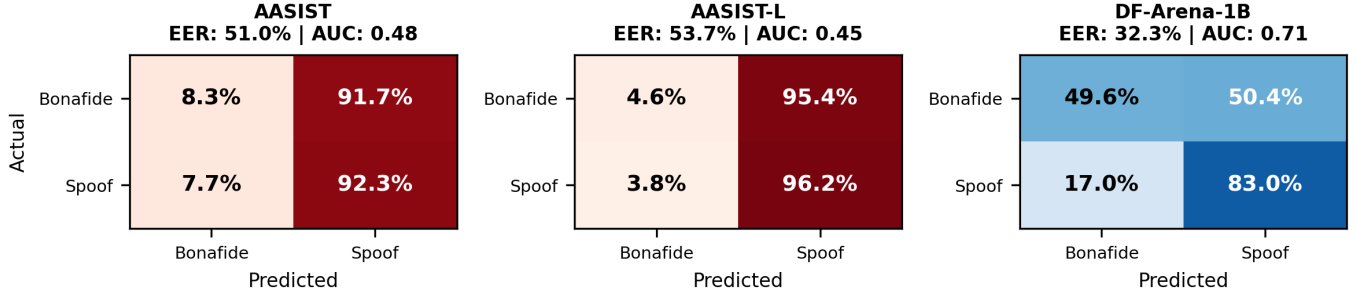


Fig. 1. Confusion matrices for all detectors. AASIST and AASIST-L classify nearly all samples as spoof regardless of true label. DF-Arena-1B shows better discrimination but still exhibits 50% FPR on bona fide speech.

The robustness set contains 6,000 enhanced bona fide recordings, 60,000 babble-noise variants, and 38,400 codec-derived variants. The codec conditions comprise 3,200 bona fide and 35,200 spoofed files. Robustness variants are not pooled into the headline overall metrics, because their unequal distribution would cause babble and codec conditions to dominate the aggregate results.

For the core set, we report Equal Error Rate (EER), Area Under the ROC Curve (AUC), macro-F1, accuracy, and spoof-class precision and recall. Robustness results are reported separately using matched clean and perturbed files. All derived variants retain the identifier of their source utterance and are not interpreted as independent source recordings.

IV. RESULTS

Table III presents the overall performance of each detector on ParlaSpoof-BR. All three obtain substantially higher EERs than those reported on ASVspoof 2019 (0.83% for AASIST, 0.99% for AASIST-L, and 1.14% for DF-Arena-1B), consistent with prior evidence of limited cross-domain generalization [11].

TABLE III
OVERALL DETECTION PERFORMANCE ON PARLASPOOF-BR

Detector	EER (%)↓	AUC↑	Prec.	Rec.	Macro-F1	Acc. (%)
AASIST	50.98	0.481	0.911	0.923	0.503	84.85
AASIST-L	53.70	0.445	0.911	0.962	0.500	87.99
DF-Arena-1B	32.30	0.715	0.944	0.830	0.595	80.05

AASIST and AASIST-L obtain EERs of 50.98% and 53.70%, with AUCs of 0.481 and 0.445, respectively, indicating limited discrimination on ParlaSpoof-BR. The full AASIST model performs only slightly better than its lightweight variant, suggesting that greater capacity within this architecture does not resolve the domain mismatch [40]. DF-Arena-1B performs better, with an EER of 32.30% and an AUC of 0.715, but remains substantially below its reported ASVspoof performance. As shown in Figure 1, AASIST and AASIST-L classify most bona fide recordings as spoofed, with false-positive rates of 91.7% and 95.4%, respectively. DF-Arena-

1B provides a more balanced result but still misclassifies approximately half of the bona fide samples.

A. Performance by Synthesis Method

Table IV reports DF-Arena-1B performance by synthesis method, including unweighted means for each attack family. On average, the detector performs better on VC than on TTS, with a lower EER (26.8% versus 39.3%), higher AUC (0.795 versus 0.633), and higher recall (94.9% versus 70.5%). However, the substantial variation within both families indicates that performance depends strongly on the individual generator.

OpenVoice-v2 obtains the lowest EER overall (19.6%), whereas VoxCPM2 and Qwen3-TTS produce the weakest results. Their recalls of 41.4% and 31.2% correspond to miss rates of 58.6% and 68.8%, respectively. Their AUC values below 0.5 indicate that the detector ranks these subsets poorly relative to the ground-truth labels. Although these results are consistent with previously reported cross-lingual limitations [17, 18], the present experiment cannot distinguish language mismatch from other generator-specific factors.

TABLE IV
DF-Arena-1B PERFORMANCE BY SYNTHESIS METHOD. MEAN ROWS REPORT UNWEIGHTED AVERAGES ACROSS GENERATORS.

Type	Method	EER (%)↓	AUC↑	Rec. (%)↑
VC	OpenVoice-v2	19.6	0.892	99.7
	kNN-VC	21.4	0.849	99.4
	X-VC	23.2	0.820	98.1
	Seed-VC	30.0	0.743	94.0
	EZ-VC	36.0	0.671	83.1
	Mean	26.8	0.795	94.9
TTS	XTTS-v2	25.5	0.808	95.8
	Chatterbox	26.1	0.799	96.5
	OmniVoice	33.9	0.705	87.5
	VoxCPM2	53.1	0.463	41.4
	Qwen3-TTS	58.0	0.390	31.2
	Mean	39.3	0.633	70.5

B. Bias Analysis

Table V ranks bias factors by their impact on DF-Arena-1B detection. For each factor, the gap is computed as the

difference between the highest and lowest recall (for methodological factors) or EER (for demographic factors) observed across levels of that dimension.

TABLE V

BIAS FACTORS RANKED BY IMPACT ON DF-ARENA-1B DETECTION. GAP IS THE MAXIMUM PERFORMANCE DIFFERENCE WITHIN EACH DIMENSION.

Rank	Bias Factor	Gap (pp)
1	Synthesis Model	68.5
2	Partial Manipulation (infill.)	44.3
3	Speaker Similarity	17.9
4	UTMOS Quality	13.2
5	SNR Level	11.1
6	Region	3.7
7	Gender	0.7

Demographic bias is minimal but present: Gender gaps are under 0.7 pp EER across all detectors. Regional variation is more substantial: DF-Arena-1B shows a 3.7 pp gap (“Norte” 30.0% vs “Sul” 33.7%), and AASIST exhibits a 7.5 pp gap. The ranking is consistent—“Norte” achieves the lowest EER while “Sul” performs worst—suggesting regional acoustic patterns (prosodic rhythm, vowel reduction) correlate with detection cues. However, these demographic disparities are dwarfed by methodological factors.

Synthesis model bias dominates: DF-Arena-1B recall ranges from 31.2% (Qwen3-TTS) to 99.7% (OpenVoice-v2)—a 68.5 pp gap (Table IV). Portuguese-optimized TTS systems evade detection at rates exceeding 60%, while VC methods remain detectable above 83%. Counter-intuitively, higher-quality synthesis is harder to detect: recall drops from 95.8% for low-similarity clones to 77.9% for high-similarity ones (18 pp gap), suggesting detection will become harder as synthesis improves.

Partial manipulation is the most effective attack: Full synthesis is unnecessary when replacing a few words suffices to invert meaning. We test whether partial manipulation can evade detection more effectively than synthesizing the entire utterance. Table VI presents the results.

TABLE VI

SPEECH INFILLING DETECTION BY PERCENTAGE AND STRATEGY. RECALL (%)—LOWER VALUES INDICATE BETTER EVASION. WORST DETECTION RATES IN **BOLD**.

Strategy	% Mod.	AASIST	AASIST-L	DF-Arena
Contiguous	25%	88.7	98.4	29.2
	50%	89.5	97.8	45.2
	75%	86.0	94.8	73.5
LLM Attack	—	86.8	98.0	29.5

The central question is whether manipulating only a single word can evade detection more effectively than synthesizing the entire utterance. The answer is yes: DF-Arena-1B achieves only 29.2% recall on 25%-modified audio, compared to 31.2% on fully synthetic Qwen3-TTS and 41.4% on VoxCPM2. That is, *less* manipulation produces *better* evasion—fewer synthetic frames mean fewer artifacts for detectors to exploit.

Detection improves monotonically with manipulation extent: recall rises from 29.2% at 25% to 45.2% at 50% to 73.5%

at 75%. This confirms that DF-Arena-1B relies on cumulative artifact density rather than boundary or transition-region detection. The LLM-guided semantic attack (29.5% recall) achieves evasion comparable to 25% resynthesis, consistent with LLM-selected edits targeting short, high-impact segments.

The practical implication is severe: an adversary need only replace a few politically consequential words—a negation, a number, a name—to invert meaning while evading the best available detector with 70% probability.

Noise injection aids evasion: TTS outputs are unnaturally clean; adding authentic babble noise masks this. At SNR 10 dB, 22.2% of previously detected deepfakes escape DF-Arena-1B detection.

Audio enhancement causes false positives: Resemble Enhance causes DF-Arena-1B to flag 98.9% of *genuine* parliamentary recordings as fake—detectors confuse neural vocoder signatures with synthesis artifacts.

Codec compression asymmetry: MP3 and OGG compression affect detection differently. On synthetic audio, both codecs preserve detectability (93.2% and 99.2% recall respectively after roundtrip). However, on *genuine* recordings, MP3 compression causes DF-Arena-1B to misclassify 18.7% as fake, while OGG causes a catastrophic **94.8%** false positive rate. OGG compression artifacts more closely resemble synthesis signatures than MP3 artifacts—a critical consideration for real-world deployment where archived audio may have undergone lossy compression.

Summary: Attack methodology dominates over demographics: synthesis model choice (68.5 pp gap), infilling percentage (44.3 pp), and attack strategy (19.7 pp) far exceed gender (0.7 pp) and regional (3.7 pp) disparities. An adversary using Portuguese-optimized TTS with 25% partial manipulation would evade DF-Arena-1B with over 70% probability. Beyond evasion, codec compression poses deployment risks: OGG roundtrip causes 94.8% FPR on genuine audio.

V. CONCLUSION

We introduced ParlaSpoof-BR, an audio deepfake benchmark constructed from 2,000 utterances by 40 regionally and gender-balanced speakers, comprising 134,400 files (11,200 bona fide, 123,200 spoofed). **Key findings:** (1) Both AASIST variants generalize poorly, with increased capacity providing only modest EER reduction. (2) Multilingual TTS models (Qwen3-TTS, VoxCPM2) yield miss rates of 68.8% and 58.6% against DF-Arena-1B. (3) Shorter partial manipulations are harder to detect: recall rises from 29.2% at 25% modification to 73.5% at 75%. (4) Babble noise further reduces robustness, causing 22.2% of previously detected samples to evade detection at 10 dB SNR. **Implications:** The dominant bias factors are methodological, not demographic: synthesis model choice (68.5 pp gap) and manipulation percentage (44.3 pp) far exceed gender (0.7 pp) and regional (3.7 pp) disparities. **Future work:** We identify three priorities: bias-aware training, domain adaptation for Portuguese political speech, and detection approaches that do not rely on cumulative artifact

density. We also plan to release a large-scale training corpus to enable domain-specific detector development.

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